



Sequence & Series

"1729 is a very interesting number; it is the smallest number expressible as the sum of two cubes in two different ways." S.Ramanujan

Sequence :

A sequence is a function whose domain is the set N of natural numbers. Since the domain for every sequence is the set N of natural numbers, therefore a sequence is represented by its range. If $f : N \rightarrow R$, then $f(n) = t_n$, $n \in N$ is called a sequence and is denoted by $\{f(1), f(2), f(3), \dots\} = \{t_1, t_2, t_3, \dots\} = \{t_n\}$

Real sequence :

A sequence whose range is a subset of R is called a real sequence.

- e.g. (i) 2, 5, 8, 11,
(ii) 4, 1, -2, -5,

Types of sequence :

On the basis of the number of terms there are two types of sequence.

- (i) Finite sequences : A sequence is said to be finite if it has finite number of terms.
(ii) Infinite sequences : A sequence is said to be infinite if it has infinitely many terms.

Example # 1 : Write down the sequence whose n^{th} term is $\frac{(-2)^n}{(-1)^n + 2}$

Solution : Let $t_n = \frac{(-2)^n}{(-1)^n + 2}$
put $n = 1, 2, 3, 4, \dots$ we get
 $t_1 = -2, t_2 = \frac{4}{3}, t_3 = -8, t_4 = \frac{16}{3}$
so the sequence is $-2, \frac{4}{3}, -8, \frac{16}{3}, \dots$

Series :

By adding or subtracting the terms of a sequence, we get an expression which is called a series. If $a_1, a_2, a_3, \dots, a_n$ is a sequence, then the expression $a_1 + a_2 + a_3 + \dots + a_n$ is a series.

- e.g. (i) $1 + 2 + 3 + 4 + \dots + n$
(ii) $2 + 4 + 8 + 16 + \dots$
(iii) $-1 + 3 - 9 + 27 - \dots$

Progression :

The word progression refers to sequence or series – finite or infinite

Arithmetic progression (A.P.) :

A.P. is a sequence whose successive terms are obtained by adding a fixed number 'd' to the preceding terms. This fixed number 'd' is called the common difference. If a is the first term & d the common difference, then A.P. can be written as $a, a + d, a + 2d, \dots, a + (n - 1)d, \dots$

e.g. $-4, -1, 2, 5, \dots$

n^{th} term of an A.P. :

Let 'a' be the first term and 'd' be the common difference of an A.P., then
 $t_n = a + (n - 1)d$, where $d = t_n - t_{n-1}$

Example # 2 : Find the number of terms in the sequence 4, 7, 10, 13,, 82.

Solution : Let a be the first term and d be the common difference
 $a = 4, d = 3$ so $82 = 4 + (n - 1)3$
 $\Rightarrow n = 27$



The sum of first n terms of an A.P. :

If a is first term and d is common difference, then sum of the first n terms of AP is

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$= \frac{n}{2} [a + \ell] \equiv nt_{\left(\frac{n+1}{2}\right)}, \text{ for } n \text{ is odd. (Where } \ell \text{ is the last term and } t_{\left(\frac{n+1}{2}\right)} \text{ is the middle term.)}$$

Note : For any sequence $\{t_n\}$, whose sum of first r terms is S_r , r^{th} term, $t_r = S_r - S_{r-1}$.

Example # 3 : If in an A.P., 3rd term is 18 and 7 term is 30, then find sum of its first 17 terms

Solution : Let a be the first term and d be the common difference

$$a + 2d = 18$$

$$a + 6d = 30$$

$$d = 3, a = 12$$

$$S_{17} = \frac{17}{2} [2 \times 12 + 16 \times 3] = 612$$

Example # 4 : Find the sum of all odd numbers between 1 and 1000 which are divisible by 3

Solution : Odd numbers between 1 and 1000 are

3, 5, 7, 9, 11, 13, ----- 993, 995, 997, 999.

Those numbers which are divisible by 3 are

3, 9, 15, 21, ----- 993, 999

They form an A.P. of which $a = 3$, $d = 6$, $\ell = 999 \therefore n = 167$

$$S = \frac{n}{2} [a + \ell] = 83667$$

Example # 5 : The ratio between the sum of n term of two A.P.'s is $3n + 8 : 7n + 15$. Then find the ratio between their 12th term

Solution :
$$\frac{S_n}{S_n'} = \frac{(n/2)[2a + (n-1)d]}{(n/2)[2a' + (n-1)d']} = \frac{3n+8}{7n+15} \text{ or } \frac{a + \{(n-1)/2\}d}{a' + \{(n-1)/2\}d'} = \frac{3n+8}{7n+15} \text{ ----- (i)}$$

we have to find
$$\frac{T_{12}}{T_{12}'} = \frac{a + 11d}{a' + 11d'}$$

choosing $(n-1)/2 = 11$ or $n = 23$ in (1),

we get
$$\frac{T_{12}}{T_{12}'} = \frac{a + 11d}{a' + 11d'} = \frac{3(23) + 8}{(23) \times 7 + 15} = \frac{77}{176} = \frac{7}{16}$$

Example # 6 : If sum of n terms of a sequence is given by $S_n = 3n^2 - 4n$, find its 50th term.

Solution : Let t_n is n^{th} term of the sequence so $t_n = S_n - S_{n-1}$.

$$= 3n^2 - 4n - 3(n-1)^2 + 4(n-1) = 6n - 7$$

$$\text{so } t_{50} = 293.$$

Self practice problems :

- (1) Which term of the sequence 2005, 2000, 1995, 1990, 1985, is the first negative term
- (2) For an A.P. show that $t_m + t_{2n+m} = 2t_{m+n}$
- (3) Find the maximum sum of the A.P. $40 + 38 + 36 + 34 + 32 + \dots$
- (4) Find the sum of first 16 terms of an A.P. $a_1, a_2, a_3, \dots$
If it is known that $a_1 + a_4 + a_7 + a_{10} + a_{13} + a_{16} = 147$

Ans. (1) 403 (3) 420 (4) 392

**Remarks :**

- (i) The first term and common difference can be zero, positive or negative (or any complex number.)
- (ii) If a, b, c are in A.P. $\Rightarrow 2b = a + c$ & if a, b, c, d are in A.P. $\Rightarrow a + d = b + c$.
- (iii) Three numbers in A.P. can be taken as $a - d, a, a + d$; four numbers in A.P. can be taken as $a - 3d, a - d, a + d, a + 3d$; five numbers in A.P. are $a - 2d, a - d, a, a + d, a + 2d$; six terms in A.P. are $a - 5d, a - 3d, a - d, a + d, a + 3d, a + 5d$ etc.
- (iv) The sum of the terms of an A.P. equidistant from the beginning & end is constant and equal to the sum of first & last terms.
- (v) Any term of an A.P. (except the first) is equal to half the sum of terms which are equidistant from it. $a_n = \frac{1}{2}(a_{n-k} + a_{n+k})$, $k < n$. For $k = 1$, $a_n = \frac{1}{2}(a_{n-1} + a_{n+1})$; For $k = 2$, $a_n = \frac{1}{2}(a_{n-2} + a_{n+2})$ and so on.
- (vi) If each term of an A.P. is increased, decreased, multiplied or divided by the same non-zero number, then the resulting sequence is also an AP.
- (vii) The sum and difference of two AP's is an AP.

Example # 7 : The numbers t ($t^2 + 1$), $-\frac{t^2}{2}$ and 6 are three consecutive terms of an A.P. If t be real, then find the next two terms of A.P.

Solution : $2b = a + c \Rightarrow -t^2 = t^3 + t + 6$
 or $t^3 + t^2 + t + 6 = 0$
 or $(t + 2)(t^2 - t + 3) = 0 \quad \therefore t^2 - t + 3 \neq 0 \Rightarrow t = -2$
 the given numbers are $-10, -2, 6$
 which are in an A.P. with $d = 8$. The next two numbers are 14, 22

Example # 8 : If a_1, a_2, a_3, a_4, a_5 are in A.P. with common difference $\neq 0$, then find the value of $\sum_{i=1}^5 a_i$, when $a_3 = 2$.

Solution : As a_1, a_2, a_3, a_4, a_5 are in A.P., we have $a_1 + a_5 = a_2 + a_4 = 2a_3$.
 Hence $\sum_{i=1}^5 a_i = 10$.

Example # 9 : If $a(b + c), b(c + a), c(a + b)$ are in A.P., prove that $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are also in A.P.

Solution : $\because a(b + c), b(c + a), c(a + b)$ are in A.P. $\Rightarrow$ subtract $ab + bc + ca$ from each
 $-bc, -ca, -ab$ are in A.P.
 divide by $-abc$
 $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P.

Example # 10 : If $\frac{a+b}{1-ab}, b, \frac{b+c}{1-bc}$ are in A.P. then prove that $\frac{1}{a}, b, \frac{1}{c}$ are in A.P.

Solution : $\therefore \frac{a+b}{1-ab}, b, \frac{b+c}{1-bc}$ are in A.P.
 $b - \frac{a+b}{1-ab} = \frac{b+c}{1-bc} - b$
 $\frac{-a(b^2 + 1)}{1-ab} = \frac{c(1+b^2)}{1-bc}$
 $\Rightarrow -a + abc = c - abc$
 $a + c = 2abc$
 divide by ac
 $\frac{1}{c} + \frac{1}{a} = 2b \Rightarrow \frac{1}{a}, b, \frac{1}{c}$ are in A.P.



Arithmetic mean (mean or average) (A.M.) :

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if a, b, c are in A.P., b is A.M. of a & c .

A.M. for any n numbers $a_1, a_2, \dots, a_n$ is; $A = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$.

n -Arithmetic means between two numbers :

If a, b are any two given numbers & $a, A_1, A_2, \dots, A_n, b$ are in A.P., then $A_1, A_2, \dots, A_n$ are the n A.M.'s between a & b .

$$A_1 = a + \frac{b-a}{n+1}, A_2 = a + \frac{2(b-a)}{n+1}, \dots, A_n = a + \frac{n(b-a)}{n+1}$$

Note : Sum of n A.M.'s inserted between a & b is equal to n times the single A.M. between a & b

$$\text{i.e. } \sum_{r=1}^n A_r = nA, \text{ where } A \text{ is the single A.M. between } a \text{ & } b \quad \text{i.e. } A = \frac{a+b}{2}$$

Example # 11 : If a, b, c, d, e, f are A. M's between 2 and 12, then find $a + b + c + d + e + f$.

Solution : Sum of A.M.'s = 6 single A.M. = $\frac{6(2+12)}{2} = 42$

Example # 12 : Insert 10 A.M. between 3 and 80.

Solution : Here 3 is the first term and 80 is the 12th term of A.P. so $80 = 3 + (11)d$

$$\Rightarrow d = 7$$

so the series is 3, 10, 17, 24,, 73, 80

$\therefore$ required means are 10, 17, 24,, 73.

Self practice problems :

(5) There are n A.M.'s between 3 and 29 such that 6th mean : $(n-1)$ th mean : : 3 : 5 then find the value of n .

(6) For what value of n , $\frac{a^{n+3} + b^{n+3}}{a^{n+2} + b^{n+2}}$, $a \neq b$ is the A.M. of a and b .

Ans. (5) $n = 12$ (6) $n = -2$

Geometric progression (G.P.) :

G.P. is a sequence of numbers whose first term is non zero & each of the succeeding terms is equal to the preceding terms multiplied by a constant. Thus in a G.P. the ratio of successive terms is constant. This constant factor is called the **common ratio** of the series & is obtained by dividing any term by that which immediately precedes it. Therefore $a, ar, ar^2, ar^3, ar^4, \dots$ is a G.P. with ' a ' as the first term & ' r ' as common ratio.

$$\text{e.g. (i) } 2, 4, 8, 16, \dots \quad \text{(ii) } \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, \dots$$

Results : (i) n^{th} term of GP = $a r^{n-1}$

(ii) Sum of the first n terms of GP

$$S_n = \begin{cases} \frac{a(r^n - 1)}{r - 1}, & r \neq 1 \\ na, & r = 1 \end{cases}$$

(iii) Sum of an infinite terms of GP when $|r| < 1$. When $n \rightarrow \infty$, $r^n \rightarrow 0$ if $|r| < 1$ therefore,

$$S_\infty = \frac{a}{1-r} \quad (|r| < 1)$$



Example # 13 : The n^{th} term of the series $3, \sqrt{3}, 1, \dots$ is $\frac{1}{243}$, then find n

Solution : $3 \cdot \left(\frac{1}{\sqrt{3}}\right)^{n-1} = \frac{1}{243} \Rightarrow n = 13$

Example # 14 : The first term of an infinite G.P. is 1 and any term is equal to the sum of all the succeeding terms. Find the series.

Solution : Let the G.P. be $1, r, r^2, r^3, \dots$

given condition $\Rightarrow r = \frac{r^2}{1-r} \Rightarrow r = \frac{1}{2}$,

Hence series is $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots, \infty$

Example # 15 : In a G.P., $T_2 + T_5 = 216$ and $T_4 : T_6 = 1 : 4$ and all terms are integers, then find its first term :

Solution : $ar(1 + r^3) = 216$ and $\frac{ar^3}{ar^5} = \frac{1}{4}$
 $\Rightarrow r^2 = 4 \Rightarrow r = \pm 2$
 when $r = 2$ then $2a(9) = 216 \Rightarrow a = 12$
 when $r = -2$, then $-2a(1-8) = 216$
 $\therefore a = \frac{216}{14} = \frac{108}{7}$, which is not an integer.

Self practice problems :

- (7) Find the G.P. if the common ratio of G.P. is 3, n^{th} term is 486 and sum of first n terms is 728.
- (8) If $x, 2y, 3z$ are in A.P. where the distinct numbers x, y, z are in G.P. Then find the common ratio of G.P.
- (9) A G.P. consist of $2n$ terms. If the sum of the terms occupying the odd places is S_1 and that of the terms occupying the even places is S_2 , then find the common ratio of the progression.
- (10) If continued product of three number in G.P. is 216 and sum of there product in pairs is 156. Find the numbers.

Ans. (7) $2, 6, 18, 54, 162, 486$ (8) $\frac{1}{3}$ (9) $\frac{S_2}{S_1}$
 (10) $2, 6, 18$

Remarks :

- (i) If a, b, c are in G.P. $\Rightarrow b^2 = ac$, in general if $a_1, a_2, a_3, a_4, \dots, a_{n-1}, a_n$ are in G.P., then $a_1 a_n = a_2 a_{n-1} = a_3 a_{n-2} = \dots$
- (ii) Any three consecutive terms of a G.P. can be taken as $\frac{a}{r}, a, ar$.
- (iii) Any four consecutive terms of a G.P. can be taken as, $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$.
- (iv) If each term of a G.P. be multiplied or divided or raised to power by the same non-zero quantity, the resulting sequence is also a G.P..
- (v) If $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are two G.P.'s with common ratio r_1 and r_2 respectively, then the sequence $a_1 b_1, a_2 b_2, a_3 b_3, \dots$ is also a G.P. with common ratio $r_1 r_2$.
- (vi) If $a_1, a_2, a_3, \dots$ are in G.P. where each $a_i > 0$, then $\log a_1, \log a_2, \log a_3, \dots$ are in A.P. and its converse is also true.



Example # 16 : Three numbers form an increasing G.P. If the middle number is doubled, then the new numbers are in A.P. The common ratio of G.P. is :

Solution : Three number in G.P. are $\frac{a}{r}$, a , ar

then $\frac{a}{r}$, $2a$, ar are in A.P. as given.

$$\therefore 2(2a) = a \left(r + \frac{1}{r} \right)$$

$$\text{or } r^2 - 4r + 1 = 0$$

$$\text{or } r = 2 \pm \sqrt{3}$$

$$\text{or } r = 2 + \sqrt{3} \text{ as } r > 1 \text{ for an increasing G.P.}$$

Example # 17 : The sum of an infinite geometric progression is 2 and the sum of the geometric progression made from the cubes of this infinite series is 24. Then find its first term and common ratio :

Solution : Let a be the first term and r be the common ratio of G.P.

$$\frac{a}{1-r} = 2, \frac{a^3}{1-r^3} = 24, -1 < r < 1$$

$$\text{Solving we get } a = 3, r = -\frac{1}{2}$$

Example # 18 : Express $0.42\dot{3}$ in the form of $\frac{p}{q}$, (where $p, q \in \mathbb{I}, q \neq 0$)

$$\text{Solution : } S = \frac{4}{10} + \frac{23}{10^3} + \frac{23}{10^5} + \dots \infty = \frac{4}{10} + \frac{a}{1-r} = \frac{4}{10} + \frac{23}{990} = \frac{419}{990}$$

Example # 19 : Evaluate $9 + 99 + 999 + \dots$ upto n terms.

$$\begin{aligned} \text{Solution : } \text{Let } S &= 9 + 99 + 999 + \dots \text{ upto } n \text{ terms.} \\ &= [9 + 99 + 999 + \dots] \\ &= [(10 - 1) + (10^2 - 1) + (10^3 - 1) + \dots + \text{ upto } n \text{ terms}] \\ &= [10 + 10^2 + 10^3 + \dots + 10^n - n] = \left(\frac{10(10^n - 1)}{9} - n \right) \end{aligned}$$

Geometric means (mean proportional) (G.M.):

If a, b, c are in G.P., b is called as the G.M. of a & c .

If a and c are both positive, then $b = \sqrt{ac}$ and if a and c are both negative, then $b = -\sqrt{ac}$.

$b^2 = ac$, therefore $b = \sqrt{ac}$; $a > 0, c > 0$.

n-Geometric means between a, b :

If a, b are two given numbers & $a, G_1, G_2, \dots, G_n, b$ are in G.P.. Then

$G_1, G_2, G_3, \dots, G_n$ are n G.M.s between a & b .

$$G_1 = a(b/a)^{1/n+1}, G_2 = a(b/a)^{2/n+1}, \dots, G_n = a(b/a)^{n/n+1}$$

Note : The product of n G.M.s between a & b is equal to the n th power of the single G.M. between a & b

$$\text{i.e. } \prod_{r=1}^n G_r = (\sqrt[n]{ab})^n = G^n, \text{ where } G \text{ is the single G.M. between } a \text{ & } b.$$

Example # 20 : Between 4 and 2916 are inserted odd number $(2n + 1)$ G.M.'s. Then the $(n + 1)$ th G.M. is

Solution : $4, G_1, G_2, \dots, G_{n+1}, \dots, G_{2n+1}, 2916$

G_{n+1} will be the middle mean of $(2n + 1)$ odd means and it will be equidistant from 1st and last term

$$\therefore 4, G_{n+1}, 2916 \text{ will also be in G.P.}$$

$$\therefore G_{n+1}^2 = 4 \times 2916 = 4 \times 9 \times 324 = 4 \times 9 \times 4 \times 81$$

$$G_{n+1} = 2 \times 3 \times 2 \times 9 = 108.$$

**Self practice problems :**

- (11) Find the value of n so that $\frac{a^{n+1} + b^{n+1}}{a^n + b^n}$ may be the G.M. between a and b .
- (12) If $a = \underbrace{111 \dots 1}_{55}$, $b = 1 + 10 + 10^2 + 10^3 + 10^4$ and $c = 1 + 10^5 + 10^{10} + \dots + 10^{50}$, then prove that
- (i) 'a' is a composite number (ii) $a = bc$.

Ans. (11) $n = -\frac{1}{2}$

Harmonic progression (H.P.)

A sequence is said to be in H.P. if the reciprocals of its terms are in A.P.. If the sequence $a_1, a_2, a_3, \dots, a_n$ is in H.P. then $1/a_1, 1/a_2, \dots, 1/a_n$ is in A.P.

Note : (i) Here we do not have the formula for the sum of the n terms of an H.P.. For H.P. whose first term is a and second term is b , the n^{th} term is $t_n = \frac{ab}{b + (n-1)(a-b)}$.

(ii) If a, b, c are in H.P. $\Rightarrow b = \frac{2ac}{a+c}$ or $\frac{a}{c} = \frac{a-b}{b-c}$.

(iii) If a, b, c are in A.P. $\Rightarrow \frac{a-b}{b-c} = \frac{a}{a}$

(iv) If a, b, c are in G.P. $\Rightarrow \frac{a-b}{b-c} = \frac{a}{b}$

Harmonic mean (H.M.):

If a, b, c are in H.P., b is called as the H.M. between a & c , then $b = \frac{2ac}{a+c}$

If $a_1, a_2, \dots, a_n$ are 'n' non-zero numbers then H.M. 'H' of these numbers is given by

$$\frac{1}{H} = \frac{1}{n} \left[\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right]$$

Example # 21 : The 7th term of a H.P. is $\frac{1}{10}$ and 12th term is $\frac{1}{25}$, find the 20th term of H.P.

Solution : Let a be the first term and d be the common difference of corresponding A.P.

$$a + 6d = 10$$

$$a + 11d = 25$$

$$5d = 15$$

$$d = 3, a = -8$$

$$T_{20} = a + 19d$$

$$= -8 + 19 \times 3 = 49$$

$$20 \text{ term of H.P.} = \frac{1}{49}$$

Example # 22 : Insert 4 H.M between $\frac{3}{4}$ and $\frac{3}{19}$.

Solution : Let 'd' be the common difference of corresponding A.P..

$$\text{so } d = \frac{\frac{19}{3} - \frac{4}{3}}{5} = 1.$$

$$\therefore \frac{1}{H_1} = \frac{4}{3} + 1 = \frac{7}{3} \quad \text{or} \quad H_1 = \frac{3}{7}$$

$$\frac{1}{H_2} = \frac{4}{3} + 2 = \frac{10}{3} \quad \text{or} \quad H_2 = \frac{3}{10}$$



$$\frac{1}{H_3} = \frac{4}{3} + 3 = \frac{13}{3} \quad \text{or} \quad H_3 = \frac{3}{13}$$

$$\frac{1}{H_4} = \frac{4}{3} + 4 = \frac{16}{3} \quad \text{or} \quad H_4 = \frac{3}{16}$$

Example # 23 : Find the largest positive term of the H.P., whose first two term are $\frac{2}{5}$ and $\frac{12}{23}$.

Solution : The corresponding A.P. is $\frac{5}{2}, \frac{23}{12}, \dots$ or $\frac{30}{12}, \frac{23}{12}, \frac{16}{12}, \frac{9}{12}, \frac{2}{12}, \frac{-5}{12}, \dots$

The H.P. is $\frac{12}{30}, \frac{12}{23}, \frac{12}{16}, \frac{12}{9}, \frac{12}{2}, -\frac{12}{5}, \dots$

Largest positive term = $\frac{12}{2} = 6$

Self practice problems :

- (13) If a, b, c, d, e are five numbers such that a, b, c are in A.P., b, c, d are in G.P. and c, d, e are in H.P. prove that a, c, e are in G.P.
- (14) If the ratio of H.M. between two positive numbers 'a' and 'b' ($a > b$) is to their G.M. as 12 to 13, prove that a : b is 9 : 4.
- (15) a, b, c are in H.P. then prove that $\frac{b+a}{b-a} + \frac{b+c}{b-c} = 2$
- (16) If a, b, c, d are in H.P., then show that $ab + bc + cd = 3ad$

Arithmetico-geometric series :

A series, each term of which is formed by multiplying the corresponding terms of an A.P. & G.P. is called the Arithmetico-Geometric Series. e.g. $1 + 3x + 5x^2 + 7x^3 + \dots$

Here 1, 3, 5, are in A.P. & 1, x, x^2 , x^3 , are in G.P..

Sum of n terms of an arithmetico-geometric series:

Let $S_n = a + (a + d)r + (a + 2d)r^2 + \dots + [a + (n-1)d]r^{n-1}$, then

$$S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]r^n}{1-r}, \quad r \neq 1.$$

Sum to Infinity: If $|r| < 1$ & $n \rightarrow \infty$, then $\lim_{n \rightarrow \infty} r^n = 0$ and $\lim_{n \rightarrow \infty} n.r^n = 0$

$$\therefore S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}.$$

Example # 24 : The sum to n terms of the series $1 + 5\left(\frac{4n+1}{4n-3}\right) + 9\left(\frac{4n+1}{4n-3}\right)^2 + 13\left(\frac{4n+1}{4n-3}\right)^3 + \dots$ is .

Solution : Let $x = \frac{4n+1}{4n-3}$, then

$$1 - x = \frac{-4}{4n-3}, \quad \frac{1}{1-x} = -\frac{(4n-3)}{4}$$

$$\frac{x}{1-x} = -\frac{(4n+1)}{4}$$

$$S = 1 + 5x + 9x^2 + \dots + (4n-3)x^{n-1}$$

$$Sx = x + 5x^2 + \dots + (4n-3)x^n$$

$$S - Sx = 1 + 4x + 4x^2 + \dots + 4x^{n-1} - (4n-3)x^n.$$

$$S(1-x) = 1 + \frac{4x}{1-x} [1 - x^{n-1}] - (4n-3)x^n$$

$$S = \frac{1}{1-x} \left[1 + \frac{4x}{1-x} - \frac{4x^n}{1-x} - (4n-3)x^n \right] = -\frac{(4n-3)}{4} [1 - (4n+1) + (4n-3)x^n - (4n-3)x^n] = n(4n-3).$$



Example # 25 : Find sum to infinite terms of the series $1 + 2x + 3x^2 + 4x^3 + \dots$, $-1 < x < 1$

Solution : let $S = 1 + 2x + 3x^2 + 4x^3 + \dots$ (i)
 $xS = x + 2x^2 + 3x^3 + \dots$ (ii)
 (i) - (ii) $\Rightarrow (1 - x) S = 1 + x + x^2 + x^3 + \dots$
 or $S = \frac{1}{(1-x)^2}$

Example # 26 : Evaluate : $1^2 + 2^2x + 3^2x^2 + 4^2x^3 + \dots$ upto infinite terms for $|x| < 1$.

Solution : Let $s = 1^2 + 2^2x + 3^2x^2 + 4^2x^3 + \dots \infty$... (i)
 $xs = 1^2x + 2^2x^2 + 3^2x^3 + \dots \infty$... (ii)
 (i) - (ii)
 $(1 - x) s = 1 + 3x + 5x^2 + 7x^3 + \dots$
 $(1 - x) s = \frac{1}{1-x} + \frac{2x}{(1-x)^2}$
 $s = \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^3}$
 $s = \frac{1-x+2x}{(1-x)^3}$
 $s = \frac{1+x}{(1-x)^3}$

Self practice problems :

(17) If $4 + \frac{4+d}{5} + \frac{4+2d}{5^2} + \dots = 1$, then find d.

(18) Evaluate : $1 + 3x + 6x^2 + 10x^3 + \dots$ upto infinite term, where $|x| < 1$.

(19) Sum to n terms of the series : $1 + 2 \left(1 + \frac{1}{n}\right) + 3 \left(1 + \frac{1}{n}\right)^2 + \dots$

Ans. (17) $-\frac{64}{5}$
 (18) $\frac{1}{(1-x)^3}$
 (19) n^2

Relation between means :

- (i) If A, G, H are respectively A.M., G.M., H.M. between a & b both being positive, then $G^2 = AH$ (i.e. A, G, H are in G.P.) and $A \geq G \geq H$.

Example # 27 : The A.M. of two numbers exceeds the G.M. by 2 and the G.M. exceeds the H.M. by $\frac{8}{5}$; find the numbers.

Solution : Let the numbers be a and b, now using the relation

$$G^2 = AH = (G + 2) \left(G - \frac{8}{5}\right) \Rightarrow G = 8 \quad ; \quad A = 10$$

i.e. $ab = 64$

also $a + b = 20$

Hence the two numbers are 4 and 16.

**A.M. ≥ G.M. ≥ H.M.**

Let $a_1, a_2, a_3, \dots, a_n$ be n positive real numbers, then we define their

$$\text{A.M.} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}, \text{ their}$$

$$\text{G.M.} = (a_1 a_2 a_3 \dots a_n)^{1/n} \text{ and their}$$

$$\text{H.M.} = \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}.$$

It can be shown that $\text{A.M.} \geq \text{G.M.} \geq \text{H.M.}$ and equality holds at either places iff $a_1 = a_2 = a_3 = \dots = a_n$

Example # 28 : If $a, b, c > 0$, prove that $\frac{ab}{c^2} + \frac{bc}{a^2} + \frac{ca}{b^2} \geq 3$

Solution : Using the relation $\text{A.M.} \geq \text{G.M.}$ we have

$$\frac{\frac{ab}{c^2} + \frac{bc}{a^2} + \frac{ca}{b^2}}{3} \geq \left(\frac{ab}{c^2} \cdot \frac{bc}{a^2} \cdot \frac{ca}{b^2} \right)^{\frac{1}{3}} \Rightarrow \frac{ab}{c^2} + \frac{bc}{a^2} + \frac{ca}{b^2} \geq 3$$

Example # 29 : If $a_i > 0 \forall i = 1, 2, 3, \dots$ prove that $(a_1 + a_2 + a_3 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right) \geq n^2$

Solution : Using the relation $\text{A.M.} \geq \text{H.M.}$

$$\begin{aligned} \frac{a_1 + a_2 + a_3 + \dots + a_n}{n} &\geq \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n}} \\ \Rightarrow (a_1 + a_2 + a_3 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_n} \right) &\geq n^2 \end{aligned}$$

Example # 30 : If x, y, z are positive then prove that $(x + y)(y + z)(z + x) \left(\frac{1}{x} + \frac{1}{y} \right) \left(\frac{1}{y} + \frac{1}{z} \right) \left(\frac{1}{z} + \frac{1}{x} \right) \geq 64$

Solution : Using the relation $\text{A.M.} \geq \text{H.M.}$

$$\frac{x + y}{2} \geq \frac{2}{\frac{1}{x} + \frac{1}{y}} \Rightarrow (x + y) \left(\frac{1}{x} + \frac{1}{y} \right) \geq 4 \quad \dots (i)$$

$$\text{similarly } (y + z) \left(\frac{1}{y} + \frac{1}{z} \right) \geq 4 \quad \dots (ii)$$

$$(z + x) \geq 4 \left(\frac{1}{z} + \frac{1}{x} \right) \quad \dots (iii)$$

$$\text{by (i), (ii) \& (iii) } (x + y)(y + z)(z + x) \left(\frac{1}{x} + \frac{1}{y} \right) \left(\frac{1}{y} + \frac{1}{z} \right) \left(\frac{1}{z} + \frac{1}{x} \right) \geq 64$$

Example # 31 : If $n > 0$, prove that $2^n > 1 + n\sqrt{2^{n-1}}$

Solution : Using the relation $\text{A.M.} \geq \text{G.M.}$ on the numbers $1, 2, 2^2, 2^3, \dots, 2^{n-1}$, we have

$$\frac{1 + 2 + 2^2 + \dots + 2^{n-1}}{n} > (1 \cdot 2 \cdot 2^2 \cdot 2^3 \cdot \dots \cdot 2^{n-1})^{1/n}$$

Equality does not hold as all the numbers are not equal.

$$\Rightarrow \frac{2^n - 1}{2 - 1} > n \left(2^{\frac{(n-1)n}{2}} \right)^{\frac{1}{n}} \Rightarrow 2^n - 1 > n 2^{\frac{(n-1)}{2}}$$

$$\Rightarrow 2^n > 1 + n 2^{\frac{(n-1)}{2}}.$$



Example # 32 : If x, y, z are positive and $x + y + z = 7$ then find greatest value of $x^2 y^3 z^2$.

Solution : Using the relation A.M. $\geq$ G.M.

$$\frac{\frac{x}{2} + \frac{x}{2} + \frac{y}{3} + \frac{y}{3} + \frac{y}{3} + \frac{z}{2} + \frac{z}{2}}{7} \geq \left(\frac{x^2}{4} \cdot \frac{y^3}{27} \cdot \frac{z^2}{4} \right)^{\frac{1}{7}}$$

$$\Rightarrow 1 \geq \left(\frac{x^2}{4} \cdot \frac{y^3}{27} \cdot \frac{z^2}{4} \right)^{\frac{1}{7}} \Rightarrow 432 \geq x^2 y^3 z^2$$

Self practice problems :

(20) If a, b, c are real and distinct, then show that $a^2(1+b^2) + b^2(1+c^2) + c^2(1+a^2) > 6abc$

(21) Prove that $2 \cdot 4 \cdot 6 \cdot 8 \dots 2n < (n+1)^n$. ($n \in \mathbb{N}$)

(22) If a, b, c, d are positive real numbers prove that $\frac{bcd}{a^2} + \frac{cda}{b^2} + \frac{dab}{c^2} + \frac{abc}{d^2} > a + b + c + d$

(23) If $x^6 - 12x^5 + ax^4 + bx^3 + cx^2 + dx + 64 = 0$ has positive roots then find a, b, c, d ,

(24) If $a, b > 0$, prove that $[(1+a)(1+b)]^3 > 3^3 a^2 b^2$

Ans. (23) $a = 60, b = -160, c = 240, d = -192$

Results :

- (i) $\sum_{r=1}^n (a_r \pm b_r) = \sum_{r=1}^n a_r \pm \sum_{r=1}^n b_r$ (ii) $\sum_{r=1}^n k a_r = \sum_{r=1}^n k a_r$
- (iii) $\sum_{r=1}^n k = k + k + k + \dots n \text{ times} = nk$; where k is a constant.
- (iv) $\sum_{r=1}^n r = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$
- (v) $\sum_{r=1}^n r^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$
- (vi) $\sum_{r=1}^n r^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$

Example # 33 : Find the sum of the series to n terms whose n^{th} term is $3n + 2$.

Solution : $S_n = \sum T_n = \sum (3n + 2) = 3\sum n + \sum 2 = \frac{3(n+1)n}{2} + 2n = \frac{n}{2} (3n + 7)$

Example # 34 : $T_k = k^3 + 3^k$, then find $\sum_{k=1}^n T_k$.

Solution : $\sum_{k=1}^n T_k = \sum_{k=1}^n k^3 + \sum_{k=1}^n 3^k = \left(\frac{n(n+1)}{2} \right)^2 + \frac{3(3^n - 1)}{3 - 1} = \left(\frac{n(n+1)}{2} \right)^2 + \frac{3}{2} (3^n - 1)$



Method of difference for finding n^{th} term :

Let $u_1, u_2, u_3, \dots$ be a sequence, such that $u_2 - u_1, u_3 - u_2, \dots$ is either an A.P. or a G.P. then n^{th} term u_n of this sequence is obtained as follows

$$S = u_1 + u_2 + u_3 + \dots + u_n \quad \dots\dots\dots(i)$$

$$S = u_1 + u_2 + \dots + u_{n-1} + u_n \quad \dots\dots\dots(ii)$$

$$(i) - (ii) \Rightarrow u_n = u_1 + (u_2 - u_1) + (u_3 - u_2) + \dots + (u_n - u_{n-1})$$

Where the series $(u_2 - u_1) + (u_3 - u_2) + \dots + (u_n - u_{n-1})$ is

either in A.P. or in G.P. then we can find u_n . So sum of series $S = \sum_{r=1}^n u_r$

Note : The above method can be generalised as follows :

Let $u_1, u_2, u_3, \dots$ be a given sequence.

The first differences are $\Delta_1 u_1, \Delta_1 u_2, \Delta_1 u_3, \dots$ where $\Delta_1 u_1 = u_2 - u_1, \Delta_1 u_2 = u_3 - u_2$ etc.

The second differences are $\Delta_2 u_1, \Delta_2 u_2, \Delta_2 u_3, \dots$, where $\Delta_2 u_1 = \Delta_1 u_2 - \Delta_1 u_1, \Delta_2 u_2 = \Delta_1 u_3 - \Delta_1 u_2$ etc.

This process is continued until the k^{th} differences $\Delta_k u_1, \Delta_k u_2, \dots$ are obtained, where the k^{th} differences are all equal or they form a GP with common ratio different from 1.

Case - 1 : The k^{th} differences are all equal.

In this case the n^{th} term, u_n is given by

$u_n = a_0 n^k + a_1 n^{k-1} + \dots + a_k$, where $a_0, a_1, \dots, a_k$ are calculated by using first ' $k + 1$ ' terms of the sequence.

Case - 2 : The k^{th} differences are in GP with common ratio r ($r \neq 1$)

The n^{th} term is given by $u_n = \lambda r^{n-1} + a_0 n^{k-1} + a_1 n^{k-2} + \dots + a_{k-1}$

Example # 35 : Find the n^{th} term of the series 1, 3, 8, 16, 27, 41,

$$\text{Solution : } s = 1 + 3 + 8 + 16 + 27 + 41 + \dots T_n \quad \dots\dots(i)$$

$$s = 1 + 3 + 8 + 16 + 27 \dots T_{n-1} + T_n \quad \dots\dots(ii)$$

$$(i) - (ii)$$

$$T_n = 1 + 2 + 5 + 8 + 11 + \dots (T_n - T_{n-1})$$

$$T_n = 1 + \left(\frac{n-1}{2} \right) [2 \times 2 + (n-2)3] = \frac{1}{2} [3n^2 - 5n + 4]$$

Example # 36 : Find the sum to n terms of the series 5, 7, 13, 31, 85 +

Solution : Successive difference of terms are in G.P. with common ratio 3.

$$T_n = a(3)^{n-1} + b$$

$$a + b = 5$$

$$3a + b = 7 \Rightarrow a = 1, b = 4$$

$$T_n = 3^{n-1} + 4$$

$$S_n = \Sigma T_n = \Sigma (3^{n-1} + 4) = (1 + 3 + 3^2 + \dots + 3^{n-1}) + 4n$$

$$\frac{1}{2} [3^n + 8n - 1]$$



Method of difference for finding s_n :

If possible express r^{th} term as difference of two terms as $t_r = \pm (f(r) - f(r \pm 1))$. This can be explained with the help of examples given below.

$$\begin{aligned} t_1 &= f(1) - f(0), \\ t_2 &= f(2) - f(1), \\ &\vdots \\ t_n &= f(n) - f(n-1) \\ \Rightarrow S_n &= f(n) - f(0) \end{aligned}$$

Example # 37 : Find the sum of n-terms of the series $2.5 + 5.8 + 8.11 + \dots$

Solution : $T_r = (3r - 1)(3r + 2) = 9r^2 + 3r - 2$

$$\begin{aligned} S_n &= \sum_{r=1}^n T_r = 9 \sum_{r=1}^n r^2 + 3 \sum_{r=1}^n r - \sum_{r=1}^n 2 \\ &= 9 \left(\frac{n(n+1)(2n+1)}{6} \right) + 3 \left(\frac{n(n+1)}{2} \right) - 2n \\ &= 3n(n+1)^2 - 2n \end{aligned}$$

Example # 38 : Sum to n terms of the series $\frac{1}{(1+x)(1+3x)} + \frac{1}{(1+3x)(1+5x)} + \frac{1}{(1+5x)(1+7x)} + \dots$

Solution : Let T_r be the general term of the series

$$\begin{aligned} T_r &= \frac{1}{[1+(2r-1)x][1+(2r+1)x]} \\ \text{So } T_r &= \frac{1}{2x} \left[\frac{(1+(2r+1)x) - (1+(2r-1)x)}{(1+(2r-1)x)(1+(2r+1)x)} \right] = \left[\frac{1}{(1+(2r-1)x)} - \frac{1}{(1+(2r+1)x)} \right] \\ \therefore S_n &= \sum T_r = T_1 + T_2 + T_3 + \dots + T_n \\ &= \frac{1}{2x} \left[\frac{1}{1+x} - \frac{1}{(1+(2n+1)x)} \right] = \frac{n}{(1+x)[1+(2n+1)x]} \end{aligned}$$

Example # 39 : Sum to n terms of the series $\frac{1}{1.4.7} + \frac{1}{4.7.10} + \frac{1}{7.10.13} + \dots$

$$\begin{aligned} \text{Solution : } T_n &= \frac{1}{(3n-2)(3n+1)(3n+4)} = \frac{1}{6} \left[\frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)} \right] \\ &= \frac{1}{6} \left[\left(\frac{1}{1.4} - \frac{1}{4.7} \right) + \left(\frac{1}{4.7} - \frac{1}{7.10} \right) + \dots + \left(\frac{1}{(3n-2)(3n+1)} - \frac{1}{(3n+1)(3n+4)} \right) \right] \\ &= \frac{1}{6} \left[\frac{1}{4} - \frac{1}{(3n+1)(3n+4)} \right] \end{aligned}$$



Example # 40 : Find the general term and sum of n terms of the series

$$1 + 5 + 19 + 49 + 101 + 181 + 295 + \dots$$

Solution : The sequence of difference between successive term 4, 14, 30, 52, 80

The sequence of the second order difference is 10, 16, 22, 28, clearly it is an A.P>

so let nth term

$$T_n = an^3 + bn^2 + cn + d$$

$$a + b + c + d = 1 \quad \dots(i)$$

$$8a + 4b + 2c + d = 5 \quad \dots(ii)$$

$$27a + 9b + 3c + d = 19 \quad \dots(iii)$$

$$64a + 16b + 4c + d = 49 \quad \dots(iv)$$

from (i), (ii), (iii) & (iv)

$$a = 1, b = -1, c = 0, d = 1 \quad \Rightarrow \quad T_n = n^3 - n^2 + 1$$

$$s_n = \sum (n^3 - n^2 + 1) = \left(\frac{n(n+1)}{2} \right)^2 - \frac{n(n+1)(2n+1)}{6} + n = \frac{n(n^2-1)(3n+2)}{12} + n$$

Self practice problems :

(25) Sum to n terms the following series

$$(i) \quad \frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \dots$$

$$(ii) \quad 1 + (1 + 2) + (1 + 2 + 3) + (1 + 2 + 3 + 4) + \dots$$

$$(iii) \quad 4 + 14 + 30 + 52 + 82 + 114 + \dots$$

(26) If $\sum_{r=1}^n T_r = (n+1)(n+2)(n+3)$ then find $\sum_{r=1}^n \frac{1}{T_r}$

Ans. (25) (i) $\frac{2n+n^2}{(n+1)^2}$ (ii) $\frac{n(n+1)(n+2)}{6}$ (iii) $n(n+1)^2$ (26) $\frac{n}{6(n+2)}$